DO NOW — RECALL

Lesson 046 · Objects in Arrays and Lists

Name: _____ Class: _____ Date: _____

1. What is a constructor used for?

2. What does private mean for an instance field?

3. What index identifies the first C# array item?

4. A rising-edge D-type flip-flop starts with Q=0. D=1 at the first rising edge. D then falls to 0 before the falling edge and stays 0. Trace Q immediately after the first rising edge, the falling edge and the next rising edge.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Initialising an object's state.

2. Model answer

It cannot be accessed directly from outside the class.

3. Model answer

Zero.

4. Model answer

1, 1, 0. Q samples D only at each rising edge and holds its value between those edges.

By the end, you can…

• A class describes an object; the array stores references to two Book instances.

• Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

• An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

Vocabulary — know these terms

object — Instance of a class.

reference — Value identifying an object instance.

aliasing — Several references to the same object.

index — Position in a sequence.

NEW LEARNING

Read, annotate and explain

Observe

A class describes an object; the array stores references to two Book instances.

Explain

Indexes start at 0. shelf[1] identifies the second reference, then .Title reads its property.

Apply

An array has a fixed length after creation; individual array elements can refer to objects whose properties change.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace the title stored at books[0] in a two-object Book array.

Trace and justify

Index 0 refers to the first object; its Title is Ada.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Given shelf titles Dune,Holes, give output of shelf[1].Title and reason. [4]

Question 2

Write C# to create an array of two Book objects with titles A and B, then print first title. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Two array elements reference the SAME Book object. Changing its title via first reference is then printed via second. Explain. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Updating a property through one reference can be visible through another alias.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.